Multivariate Statistical Analysis of International Athletics Performance

Thomas Barnes

1 Introduction

Within this report we will be analysing a dataset containing national record times for men's track events from 55 countries immediately prior to the 1984 Olympic Games. We have results for the 100m, 200m, 400m and 800m races. Upon initial analysis of our dataset, we can see that the measurement for the 800m race has been measured in minutes, as opposed to seconds, so we need to first convert that A.1 before continuing with our analysis. Once this has been changed, we can view some initial summary statistics for the data, as shown in 1 below A.2.

	m100	**m200**	**m400**	**m800**
Min	9.93	19.72	43.86	102.0
1st Quartile	10.27	20.59	45.56	105.3
Median	10.41	20.81	46.10	107.4
Mean	10.47	20.94	46.44	107.6
3rd Quartile	10.59	21.29	47.30	108.9
IQR	0.32	0.69	1.74	3.60
Max	12.18	23.20	52.94	121.2

Table 1: Summary statistics

These summary statistics highlight the differences in times between each event. As expected, both the mean and median values increase as the distance of the race increases. The interquartile range (IQR), which measures variability, also increases with distance from 0.32 seconds in the 100m to 3.60 seconds in the 800m, reflecting greater performance differences in longer races. This could be due to differences for example pacing strategy or fatigue in longer races, which do not need to occur in shorter races like the 100m.
To gain deeper insight into how the performance and variability scales with race distance, we can compare the scaled means and IQRs by standardising the times across events, multiplying the m100 times by 8, the m200 times by 4, and the m400 times by 2, allowing for a direct comparison to the m800 race.

Scaled Metrics	**m100**	**m200**	**m400**	**m800**
Mean	83.76873	83.76145	92.87745	107.59636
IQR	2.56	2.76	3.48	3.60

Table 2: Scaled statistics

The scaled means for the 100m and 200m races are nearly identical, indicating a consistent relationship between shorter sprinting events. Similarly the interquartile ranges are also similar for the two events, demonstrating low deviation in the shorter sprinting events. This is in contrast to the longer 400m and 800m races, at these distances, the scaled mean times start to increase quite dramatically per unit distance, as suggested earlier this may be due to increased fatigue for longer events. Examining the scaled interquartile ranges for these longer events, we can see that these also increase, suggesting for a longer race we have greater variability of time per unit distance.

2 Exploratory Analysis and Normality Assessment

We will first examine the distribution of race results for each event, to analyse their normality. This is necessary, to be able to formulate further hypothesis tests in the future. We will start by examining the scatter plots for each of our variables with one another A.3, on top of these data points we have included an ellipse. This ellipse represents a 95% data ellipse under the assumption of bivariate normality, generated from both the means and the covariance matrices for each pair of races. Upon inspection of the scatter plot, we can see that the majority of the data does lie within the 95% data ellipse, suggesting that the assumption of bivariate normality is broadly reasonable, although this does not confirm normality. For similar races for example the 100m vs 200m and the 200m vs 400m races, we have tight clustering within the ellipse, as compared to races with greater distance difference like the 100m vs 800m and 200m vs 800m. The data in these cases is still largely within our 95% data ellipse, however the datapoints are more dispersed.

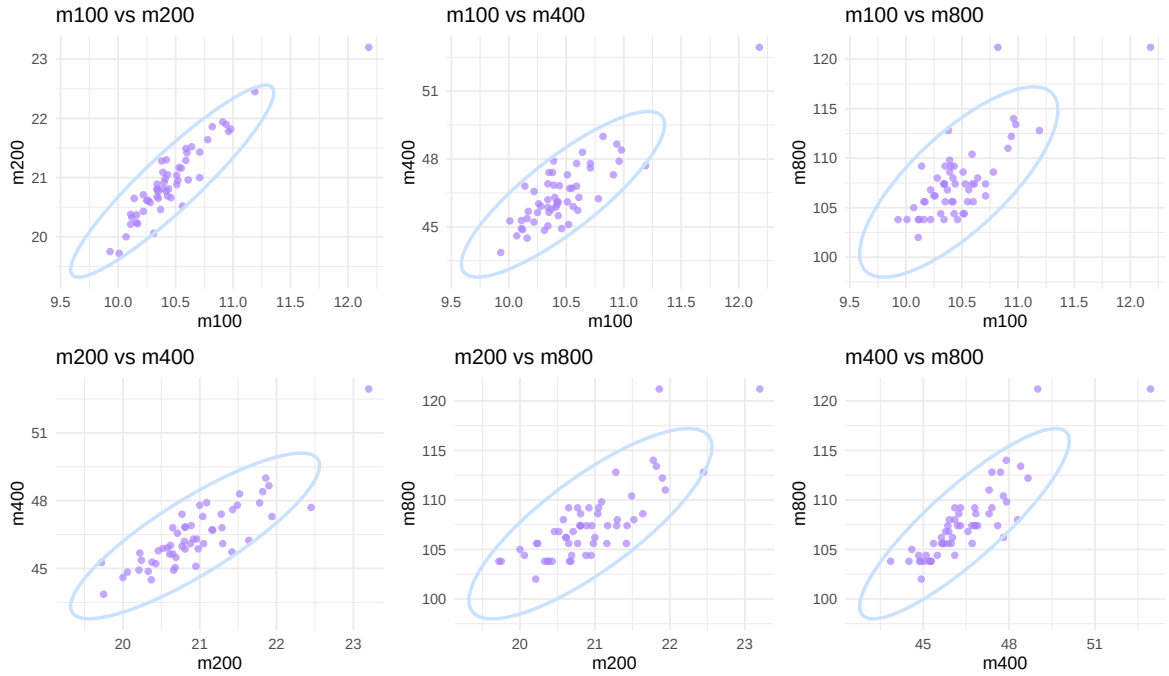


Figure 1: Pairwise scatterplots with 95% normal ellipses

From the figure, we can suggest that the assumption of bivariate normality is reasonable for each pair of races, although this cannot be confirmed from the plots alone. However, we cannot take for fact the idea that this is either trivariate normally distributed or normally distributed in 4 dimensions. If data is normally distributed in p dimensions, it is normally distributed in q dimensions for $q < p$ [1], we can conclude that univariate normality is reasonable due to this fact. Normality is not necessarily true for $q > p$, as adding new variables does not guarantee that their joint distribution with the existing variables is multivariate normal. For the purposes of this report, we will assume the data is multivariate normal in 4 dimensions, as future methodology relies on this fact.

3 Relationship Between Race Distance and Performance

We want to analyse if our race time is proportional with the distance of the race, that is we want to test if $\mu_{800} = 2\mu_{400} = 4\mu_{200} = 8\mu_{100}$ where μ_i denotes the mean time of a race of i meters. Essentially we are testing if the average speed of each race is equal.

Hypotheses:
H_0 : $\mu_{800} = 2\mu_{400} = 4\mu_{200} = 8\mu_{100}$
H_1 : At least one of these equalities does not hold (average speeds differ between events).

Before we begin our statistical test, we can make a linear transformation $Y = XA^T$, where X is our data matrix and A is our transformation matrix, such that if $\mu_{800} = 2\mu_{400} = 4\mu_{200} = 8\mu_{100}$, the mean vector $\bar{\boldsymbol{y}}$ of data matrix Y is equal to $\boldsymbol{\mu}_0 = \underline{\mathbf{0}}$. We can construct this matrix A as:

$$A = \begin{pmatrix} 0 & 0 & -2 & 1 \\ 0 & -4 & 2 & 0 \\ -8 & 4 & 0 & 0 \end{pmatrix}$$

For this we will perform a one-sample Hotelling's test, using F-tables to calculate our p value.

3.1 One Sample Hotelling's Test

We have our Hotelling's T^2 statistic $T^2 = n(\bar{\boldsymbol{y}} - \boldsymbol{\mu}_0)^T S^{-1}(\bar{\boldsymbol{y}} - \boldsymbol{\mu}_0)$. In this case, since we have transformed our data matrix X such that $Y = XA^T$, H_0 is true when the column means of $Y = \underline{\mathbf{0}}$. Thus we have $\boldsymbol{\mu}_0 = \underline{\mathbf{0}}$, giving a new test statistic:

$$T^2 = n\bar{\boldsymbol{y}}^T S^{-1}\bar{\boldsymbol{y}}$$

Now we have our test statistic T^2, we can relate it to the F-test by the following relation:

$$\left(\frac{n-p}{p(n-1)}\right) T^2(p, n-1) = F(p, n-p)$$

The calculated F-statistic provides the basis for computing the p-value, which is determined using both the test statistic and the associated degrees of freedom. This gives a p-value A.4 of $5.494444 \times 10^{-53} \ll 0.05$ and thus we can reject the hypothesis $H_0 : \ \mu_{800} = 2\mu_{400} = 4\mu_{200} = 8\mu_{100}$.

Although the Hotelling's Test has rejected our hypothesis H_0 that average speed was equal between all 4 races, it does not specify which of these races caused this violation. This prompts our next investigation into deriving simultaneous confidence intervals. We have simultaneous confidence intervals in this case in the form:

$$\boldsymbol{a}^T\bar{\boldsymbol{y}} \pm \sqrt{T^2_{p,n-1}(1-\alpha)n^{-1}\boldsymbol{a}^T S\boldsymbol{a}}$$

where $T^2_{p,n-1}(1-\alpha)$ denotes the $(1-\alpha)$ quantile of the $T^2(p, n-1)$ distribution. Equivalently:

$$\boldsymbol{a}^T\bar{\boldsymbol{y}} \pm \sqrt{\left(\frac{p(n-1)}{n-p}\right) F_{p,n-p}(1-\alpha)n^{-1}\boldsymbol{a}^T S\boldsymbol{a}}$$

where once again $F_{p,n-p}(1-\alpha)$ represents the $(1-\alpha)$ quantile of the $F(p, n-p)$ distribution. In this case we will use values of $\boldsymbol{a}$ as $\boldsymbol{a}_1 = [1,0,0]^T$, $\boldsymbol{a}_2 = [0,1,0]^T$ and $\boldsymbol{a}_3 = [0,0,1]^T$. Below you will find the result of calculating these confidence intervals A.4 as well as the hypothesised value.

Vector	Lower Bound	Upper Bound	Hypothesised Value
$\boldsymbol{a}_1$	13.95381	15.48401	0
$\boldsymbol{a}_2$	8.506643	9.725357	0
$\boldsymbol{a}_3$	-0.4377822	0.4232367	0

Table 3: Simultaneous confidence intervals

From the table, we can see that two of the three equalities we were hypothesising are not true to a 5% significance level, namely $2\mu_{400} \neq \mu_{800}$ and $2\mu_{200} \neq \mu_{400}$. The results suggest that while shorter races like the 100m and 200m exhibit relatively consistent average speeds, this does not scale with longer distances. This may reflect physiological fatigue which is not necessarily present in shorter distance events.

4 Comparison of Performance by Country Classification

Within our dataset for this race, we have two categories consisting of 'developed' countries and 'less developed' countries. We wish to analyse whether this factor has any association with the performance of the countries. We want to construct tests for the following hypotheses:

$H_0 : \boldsymbol{\mu}_{\text{developed}} = \boldsymbol{\mu}_{\text{less developed}}$
$H_1 : \boldsymbol{\mu}_{\text{developed}} \neq \boldsymbol{\mu}_{\text{less developed}}$

Here, $\boldsymbol{\mu}_{\text{developed}}$ and $\boldsymbol{\mu}_{\text{less developed}}$ are the mean vectors of race times for athletes from developed and less developed countries, respectively, across the four events. Let us begin by visualising the difference in results for all 4 races A.5. Within the original dataset, status 1 represents developed countries, while status 3 represents less-developed countries.

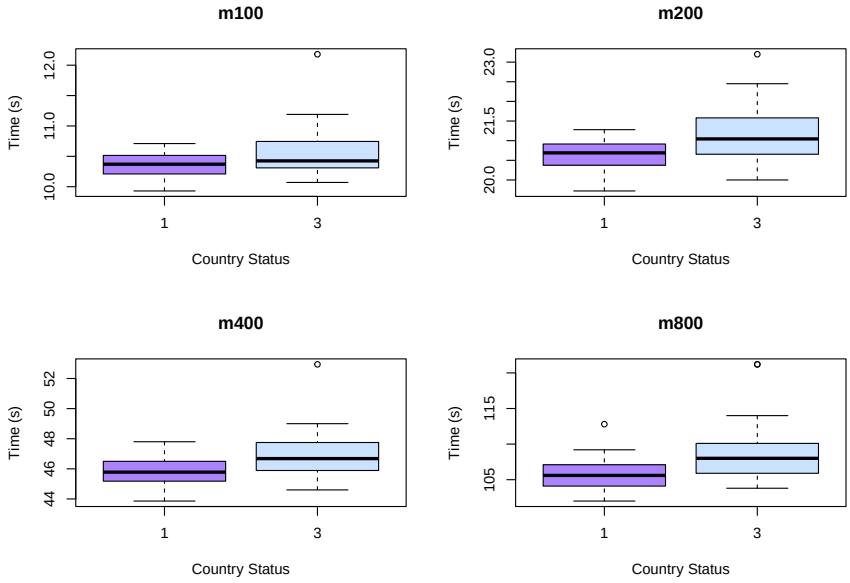


Figure 2: Comparison of times for each race

Across all distances, countries classified as 'developed' consistently exhibit better performance, as indicated by lower median times. Additionally, their performance is more consistent, with narrower interquartile ranges and fewer outliers. In comparison, 'less developed' countries exhibit slower median times as well as higher variability indicated by a greater interquartile range, alongside this there are outliers for all races as compared to just one outlier in all races for 'developed' countries. These results for poorer performance in 'less developed' countries may be associated with a difference in training, resources or other conditions which may affect an athletes performance.

4.1 Two Sample Hotelling's T^2 Test

To test whether our conclusions from visual inspection are significant, we can construct a statistical test in the form of a two sample Hotelling's T^2 test A.5. Define X_1 as our data matrix for 'developed' countries and X_2 as the data matrix for 'less developed' countries with mean vectors $\bar{\boldsymbol{x}}_1$ and $\bar{\boldsymbol{x}}_2$ respectively with sample sizes n_1 and n_2 and covariance matrices S_1 and S_2. We can create a pooled estimate of the variance matrix by:

$$S = \frac{(n_1 - 1)S_1 + (n_2 - 1)S_2}{n_1 + n_2 - 2}$$

We can form our T^2 test statistic once again by:

$$T^2 = \left(\frac{n_1 n_2}{n_1 + n_2}\right)(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)^T S^{-1}(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)$$

This yields us $T^2 = 19.76141$, as in the last section, we need to convert this into an F-statistic, in order to generate a p-value for our data. We have the relation:

$$\frac{n_1 + n_2 - p - 1}{(n_1 + n_2 - 2)p} T^2(p, n_1 + n_2 - 2) = F(p, n_1 + n_2 - p - 1)$$

We can generate a p-value using the degrees of freedom displayed, which equates to $0.0028242 < 0.05$, thus we have a significant difference between the performance of the two categories, rejecting H_0.

As in the last question, we have no significant evidence that times between 'developed' and 'less developed' are equal, indicating a significant difference (H_1). But is this the case in all races, or just one event? To investigate this we can once again construct simultaneous confidence intervals of the form:

$$\boldsymbol{a}^T(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2) \pm \sqrt{T^2_{p,n_1+n_2-2}(1-\alpha)\frac{n_1 + n_2}{n_1 n_2}\boldsymbol{a}^T S \boldsymbol{a}}$$

where S is our pooled covariance matrix as defined above and $T^2_{p,n_1+n_2-2}(1-\alpha)$ is our $(1-\alpha)$ quantile of the $T^2(p, n_1 + n_2 - 2)$ distribution, this is also equivalent to:

$$\boldsymbol{a}^T(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2) \pm \sqrt{\frac{p(n_1 + n_2 - 2)}{n_1 + n_2 - p - 1} F_{p,n_1+n_2-p-1}(1-\alpha)\frac{n_1 + n_2}{n_1 n_2}\boldsymbol{a}^T S \boldsymbol{a}}$$

where $F_{p,n_1+n_2-p-1}(1-\alpha)$ is equivalent to the $(1-\alpha)$ quantile of the $F(p, n_1+n_2-p-1)$ distribution. In this case we can set $\boldsymbol{a}$ as $\boldsymbol{a}_1 = [1,0,0,0]^T$, $\boldsymbol{a}_2 = [0,1,0,0]^T$, $\boldsymbol{a}_3 = [0,0,1,0]^T$ and $\boldsymbol{a}_4 = [0,0,0,1]^T$ to obtain confidence intervals as in the table below A.5:

Vector	Lower Bound	Upper Bound	Hypothesised $(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)$
$\boldsymbol{a}_1$	-0.5093710	0.1020884	0
$\boldsymbol{a}_2$	-1.08252024	-0.02321346	0
$\boldsymbol{a}_3$	-2.2665022	0.2128609	0
$\boldsymbol{a}_4$	-6.2899570	0.0551744	0

Table 4: SCI's for 'developed' vs 'less developed' countries

From these SCI's, we can clearly see that the only vector $\boldsymbol{a}_i$ for which the corresponding difference was significant was $\boldsymbol{a}_2$ representing the 200m race. All other races showed no statistically significant difference between the timings of 'developed' and 'less developed' countries. The significant difference in only the 200m event could be associated with specific disparities in resources, coaching, or conditions specific to this race distance. This could warrant further investigation into what these disparities are.

5 Conclusion

This report investigated 2 key questions for race results from immediately before the 1984 Olympic Games, whether race times scale proportionally with distance and whether performance differs between athletes from developed and less developed countries.

Using the one sample Hotelling's T^2 test, we analysed if there was a significant difference in average speeds between the four events analysed, to which we found there were. We then used simultaneous confidence intervals to deduce that a significant difference in speed was displayed by the 400m vs the 800m race and the 200 vs 400m race.

We then wanted to test further if the economic classification of a country has any association with its performance across each race. In this case we used a two sample Hotelling's T^2 test which did reveal

there was a significant difference between the performance of athletes. As with the last question we then proceeded to use simultaneous confidence intervals to investigate if it was all distances affected, or if it was only in one or two. Using this method we found there was only a significant difference in the performance of athletes within the 200m race.

References

[1] M. Haugh. Multivariate distributions. URL https://www.columbia.edu/~mh2078/QRM/MultivariateDistributions.pdf. Page 2, under marginal distribution header, accessed 1/12/24.

Appendices

A R Code

A.1 Importing Libraries and Data

Importing the data was completed as shown below, notably converting m800 into seconds.

```
library(ggplot2)
library(gridExtra)

data = read.table('athlrecs.txt')
data["m800"] = data["m800"] * 60
X = as.matrix(data[,1:4])
```

A.2 Summary Statistic Generation

```
summary(X)
X_IQR = apply(X, 2, IQR)

# Scaled Summary Statistics
average_X_scaled = colMeans(X) * c(8,4,2,1)
X_IQR_scaled = X_IQR * c(8,4,2,1)
```

A.3 Question 1

Code used to generate the ellipse in question 1 is as follows:

```
n_cols = length(colnames(X))
plots = list()
for (i in seq_len(n_cols - 1)) {
  for (j in seq((i + 1), n_cols)) {
    col1 = colnames(X)[i]
    col2 = colnames(X)[j]
    p = ggplot(X, aes_string(col1, col2)) +
      geom_point(alpha = 0.7, colour = "mediumpurple1") +
      labs(title = paste(col1, "vs", col2),
           x = col1,
           y = col2) +
      stat_ellipse(type = "norm",
                   level = 0.95,
                   color = "lightsteelblue1",
                   size = 1) +
      theme_minimal()
    plots[[length(plots) + 1]] = p
  }
}
do.call(grid.arrange, c(plots, ncol = 3))
```

A.4 Question 2

$Y = XA^T$ has been generated in R is follows:

```
A = matrix(c(0,0,-2,1,
             0,-4,2,0,
             -8,4,0,0), nrow = 3, byrow = T)
Y = X%*%t(A)
```

The statistical test to obtain a p-value in question 2 is as follows, we have not included a value of $\boldsymbol{\mu}_0$ as it is equal to the zero vector and thus redundant programmatically:

```
y_bar = colMeans(Y)
S = var(Y)
n = dim(Y)[1]
m = n-1
p = dim(Y)[2]
tsq = n * t(y_bar) %*% solve(S) %*% y_bar
fstat = (n-p)/(p*m) * tsq

df1 = p
df2 = n-p

p_val = pf(fstat, df1 = df1, df2 = df2, lower.tail = F)
```

The code to calculate the SCI's in question 2:

```
compute_CI = function(a, y_bar, S, p, n, df1, df2) {
  critical_val = sqrt(((p * (n-1)) / (n-p)) * qf(0.95, df1, df2) * (1 / n))
  margin = critical_val * sqrt(t(a) %*% S %*% a)
  c(t(a) %*% y_bar - margin, t(a) %*% y_bar + margin)
}

a_vectors = diag(p)
CIs = apply(a_vectors, 2, compute_CI, y_bar, S, p, n, df1, df2)
```

A.5 Question 3

The code to generate the box-plot describing the differences in 'developed' vs 'less developed' in each race is:

```
X2 = as.matrix(cbind(X, data["status"]))

par(mfrow = c(2,2))
for (col in colnames(X2[,1:4])){
  boxplot(X2[,col] ~ X2[,"status"],
          main = col,
          xlab = "Country Status",
          ylab = "Time (s)",
          col = c("mediumpurple1", "lightsteelblue1"))
}
```

The code for the statistical test present in question 3 is:

```
X1 = X2[X2[, "status"] == 1, ]
X2 = X2[X2[, "status"] == 3, ]
X1 = X1[, colnames(X2) != "status"]
X2 = X2[, colnames(X2) != "status"]


x_1bar = colMeans(X1)
x_2bar = colMeans(X2)
p = dim(X1)[2]

S_1 = cov(X1)
S_2 = cov(X2)

n_1 = dim(X1)[1]
n_2 = dim(X2)[1]


S = ((n_1 - 1)*S_1 + (n_2 - 1)*S_2)/(n_1 + n_2 - 2)

tsq = (n_1*n_2)/(n_1 + n_2) *
  t(x_1bar - x_2bar) %*% solve(S) %*% (x_1bar - x_2bar)

df1 = p
df2 = n_1 + n_2 - p - 1

fstat = (n_1 + n_2 - p - 1)/((n_1 + n_2 - 2)*p) * tsq
p_val = pf(fstat, df1 = df1, df2 = df2, lower.tail = F)
```

The code for generating the simultaneous confidence intervals in question 3 is:

```
compute_CI_diff = function(a, x_1bar, x_2bar, S, p, n_1, n_2, df1, df2) {
  critical_val = sqrt((p * (n_1 + n_2 - 2) / (n_1 + n_2 - p - 1)) *
                  qf(0.95, df1, df2) * ((n_1 + n_2) / (n_1 * n_2)))
  margin = critical_val * sqrt(t(a) %*% S %*% a)
  c(t(a) %*% (x_1bar-x_2bar) - margin, t(a) %*% (x_1bar-x_2bar) + margin)
}

a_vectors_diff = diag(p)
CIs_diff = apply(a_vectors_diff, 2, compute_CI_diff,
                 x_1bar, x_2bar, S, p, n_1,
                 n_2, df1, df2)
```